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Reducing Force Ripple of Electrostatic Film Actuators via Electrode Offset Layout

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***Abstract*—Robots driven by electromagnetic actuators are often constrained by large volume and mass, limiting their application in lightweight and compact systems. Electrostatic film actuators are lightweight, flexible, and high-precision candidates for driving miniature robotic and human–machine interactive devices. However, conventional straight electrode layouts exhibit significant force ripple (with a ripple rate of up to 1.2), leading to noticeable nonlinearity error of displacement. This work introduces an electrode offset layout that axially divides the slider electrodes into three segments with a spatial phase offset, enabling ripple suppression through force superposition. Compared with the straight layout actuator, the proposed actuator reduced the force ripple rate to 0.290 (a 53.4% reduction) and lowered the displacement nonlinearity error to 0.115% of full scale (a 13.5% reduction). The offset layout actuator achieved a maximum output force of 0.911 N and a maximum velocity of 241 mm/s, while having a flexible structure and weighing only 1.72 g. Demonstrations in a lens focusing module, an ultrathin retractable smartphone camera module, a flip camera module, and a circumferential wide-range camera module validate its applicability and potential for integration in compact optical systems.**

***Index Terms*—Electrostatic actuator, flexible actuator, soft actuator.**

I. Introduction

ROBOTS have been widely used in both industrial and social domains. Traditional robots driven by rigid electromagnetic motors often suffer from bulky volume and heavy weight [1], [2], which particularly limits their applications in miniature devices. Soft robots, typically composed of low-modulus materials or flexible thin films, provide inherent compliance [3], [4]. They have been employed in crawling robots [5], [6], grippers [7], [8], and wearable devices [9], [10]. Nevertheless, most soft robots are constrained by the inherent nonlinear characteristics of their viscoelastic materials, resulting in limited actuation control and slow response [11].

Electrostatic film actuators (EFAs) embed arrays of micron-scale electrodes into polymer films and achieve synchronous stepping motion under electrostatic fields [12]. Benefiting from their high open-loop positioning accuracy, lightweight construction, flexibility, and low power consumption, EFAs have attracted increasing attention. In particular, they have been explored for applications in human–machine interfaces [13], [14], surgical devices [12], [15], and thin robots [16], [17], [18]. However, most EFAs adopt a parallel straight electrode layout [17], [18], [19], in which the actuation force periodically varies with the relative displacement between slider and stator. This variation leads to significant force ripple, with a ripple rate up to 1.2 [20]. The ripple induces stepwise motion and vibrations of the actuator [20], thereby increasing the nonlinearity error of displacement. Such nonlinearity constrains the use of EFAs in applications requiring smooth motion, such as precision optical devices. Approaches in permanent magnet linear actuators suppress force ripple through independent-coil drive [21], mover skew optimization [22], or segmented-primary designs [23]. Although electrode skew layout [20], [24], [25] has been proposed for EFAs to reduce the theoretical ripple rate to approximately 0.18, it leads to reduced output force compared with conventional straight layouts [20].

The contributions of this work are summarized as follows. We propose an EFA with a novel electrode offset layout, as illustrated in Fig. 1(a) and (b). In this design, the slider electrodes are divided axially into three segments (two groups): two side segments (electrode group 1) of equal length and a

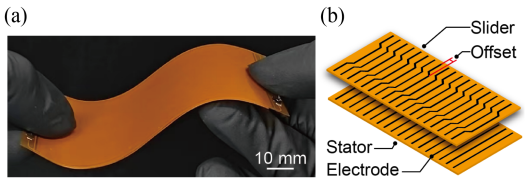


Fig. 1. Electrode offset layout EFA. (a) Flexible and thin structure. (b) Electrode design of the slider (offset layout) and stator (straight layout).

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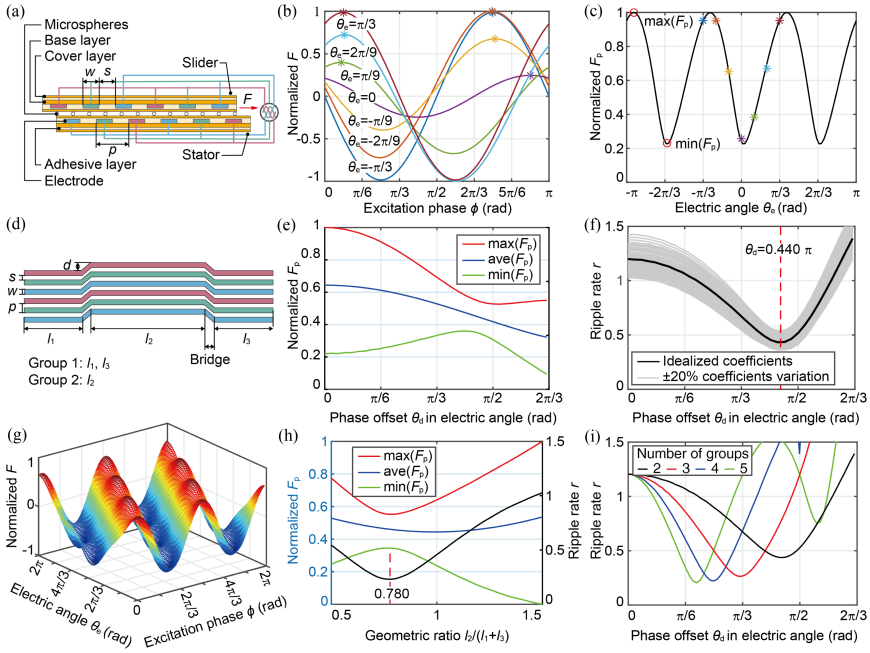


Fig. 2. Design and principle of EFAs. (a) Conventional straight electrode layout. (b) Modeled normalized actuation force versus excitation phase at different electric angles. (c) Modeled normalized peak actuation force versus electric angle. (d) Electrode design of the offset layout actuator. (e) Effect of phase offset on normalized actuation force. (f) Effect of phase offset on force ripple, with a sensitivity analysis in which the coupling coefficients C_{m1}, C_{m2}, and C_{m3} are varied within ±0%–20%. (g) Actuation force characteristics of the offset layout actuator at a phase offset of $\theta_{\mathrm{d}} = 0.440\pi$. (h) Effect of the geometric ratio $l_2/(l_1 + l_3)$ on actuation force and force ripple. (i) Effect of varying the number of electrode groups on force ripple.

central segment (electrode group 2) whose length equals the total length of the side segments. By axially offsetting the central segment relative to the side segments, spatial phase differences are introduced, enabling effective suppression of force ripple through force superposition. Compared with EFAs using the straight layout with identical electrode design parameters (pitch, width, and spacing), the offset layout reduced force ripple rate by 53.4% (0.622–0.290), enhanced output force by 22.3% (0.745–0.911 N), and lowered the nonlinearity error of displacement by 13.5% (0.133%–0.115% of full scale). Unlike the skew layout, which reduces ripple but compromises force output, the proposed offset layout simultaneously achieves low ripple and enhanced force. Leveraging these improvements, we demonstrated the actuator's potential in lens focusing, ultrathin retractable camera, flip camera, and circumferential wide-range camera modules.

The rest of this article is organized as follows. Section II presents the design, analytical model, and finite element analysis (FEA) of the offset layout EFA. Section III introduces the fabrication process of EFAs. Section IV describes the experimental setups and evaluates actuator performance. Section V demonstrates its applications in the actuation of optical devices. Finally, Section VI concludes this article and discusses future work.

II. Actuator Design and Analyses

A. Design and Principle

The driving principle and structure design of EFAs with the straight layout are illustrated in Fig. 2(a). Three-phase electrodes are arranged in parallel on the base layer and electrically insulated from the environment by an adhesive layer and a cover layer. In the electrode design parameters, w denotes the electrode width, s represents the spacing between adjacent electrodes, and p is the electrode pitch ($p = w + s$). During operation, microspheres (SLGMS-2.5, Cospheric LLC) with diameters ranging from 15 to 22 μm are uniformly distributed between the slider and the stator using a cotton swab to reduce friction. When three-phase sinusoidal alternating voltages with a phase shift of 120° are applied to the electrodes, the slider performs synchronous motion, and its velocity can be expressed as

$$v = 6pf \tag{1}$$

where f is the driving frequency of the applied voltages.

Based on the capacitance matrix model of the EFA [20], the output force of the slider can be expressed as

$$\begin{aligned} F(\phi, \theta_{\mathrm{e}}) = KV_{\mathrm{0\text{-}p}}^2 \Big[& C_{\mathrm{m1}} \sin\left(\theta_{\mathrm{e}} - 2\phi + \tfrac{\pi}{3}\right) \\ & + 2C_{\mathrm{m2}} \sin\left(2\theta_{\mathrm{e}} + 2\phi - \tfrac{\pi}{3}\right) \\ & - C_{\mathrm{m3}} \sin\left(3\theta_{\mathrm{e}}\right) \Big] \end{aligned} \tag{2}$$

where $K = kA_{\mathrm{e}}$ is the actuation force constant, with units of $\mathrm{N{\cdot}V^{-2}}$. Here, k is a material-dependent force coefficient with units of $\mathrm{N{\cdot}m^{-2}{\cdot}V^{-2}}$, and A_{e} denotes the effective actuation area. The parameter $V_{\mathrm{0\text{-}p}}$ denotes the zero-to-peak amplitude of the applied voltages. The electric angle θ_{e} is defined as $\theta_{\mathrm{e}} = 2\pi x_{\mathrm{sl}}/(3p)$, where x_{sl} is the slider displacement relative to the stator. The parameter ϕ denotes the excitation phase of the applied voltages. The coefficients C_{m1}, C_{m2}, and C_{m3} are measured parameters of the capacitance matrix. Since the actuator design parameters in [20] are similar to those used in this work, the corresponding measured values are adopted here.

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The relationships between the normalized actuation force and the excitation phase ϕ at different electric angles θ_e are shown in Fig. 2(b). The maximum value at each electric angle is defined as the peak force F_p. The relationship between the normalized peak force F_p and the electric angle θ_e is presented in Fig. 2(c). The force ripple rate is calculated as $r = (\max(F_p) - \min(F_p))/\operatorname{ave}(F_p)$.

The offset layout design is illustrated in Fig. 2(d). The electrodes are axially divided into three segments (two groups), where the length of the first segment (l_1) equals that of the third segment (l_3), and the length of the second segment is given by $l_2 = l_1 + l_3$. The electrodes of group 2 are offset by a distance d relative to group 1, and the adjacent electrode segments are connected through inclined bridges. Assuming that the force generated by the electrodes of group 1 is denoted as $F_1(\phi, \theta_e)$, and the force generated by group 2 is denoted as $F_2(\phi, \theta_e + \theta_d)$, where the phase offset is defined as $\theta_d = 2\pi d/(3p)$. Using a standard trigonometric identity, the total force generated by the actuator can be expressed as

$$\begin{aligned} F_a(\phi, \theta_e, \theta_d) &= F_1(\phi, \theta_e) + F_2(\phi, \theta_e + \theta_d) \\ &= KV_{0\text{-}p}^2 \Big\{ C_{m1} \big[\sin\big(\theta_e - 2\phi + \tfrac{\pi}{3}\big) \\ &\quad + \sin\big(\theta_e - 2\phi + \tfrac{\pi}{3} + \theta_d\big)\big] \\ &\quad + 2C_{m2}\big[\sin\big(2\theta_e + 2\phi - \tfrac{\pi}{3}\big) \\ &\quad + \sin\big(2\theta_e + 2\phi - \tfrac{\pi}{3} + 2\theta_d\big)\big] \\ &\quad - C_{m3}\big[\sin(3\theta_e) + \sin(3\theta_e + 3\theta_d)\big] \Big\} \\ &= 2KV_{0\text{-}p}^2 \big[C_{m1}\cos\big(\tfrac{\theta_d}{2}\big)\sin\big(\theta_e - 2\phi + \tfrac{\pi}{3} + \tfrac{\theta_d}{2}\big) \\ &\quad + 2C_{m2}\cos(\theta_d)\sin\big(2\theta_e + 2\phi - \tfrac{\pi}{3} + \theta_d\big) \\ &\quad - C_{m3}\cos\big(\tfrac{3\theta_d}{2}\big)\sin\big(3\theta_e + \tfrac{3\theta_d}{2}\big)\big]. \end{aligned} \tag{3}$$

Based on this equation, the relationship between the force ripple rate and the phase offset was obtained, as shown in Fig. 2(e) and (f). The ripple rate reaches its minimum at $\theta_d = 0.440\pi$, and this optimal phase-offset range remains stable under ±0%–20% variations in the coupling coefficients C_{m1}, C_{m2}, and C_{m3}. The force characteristic $F_a(\phi, \theta_e)$ is plotted in Fig. 2(g) to further illustrate the reduction in actuation force ripple. The effects of the geometric ratio $l_2/(l_1 + l_3)$ and the number of electrode groups are shown in Fig. 2(h) and (i).

Overall, with manageable wiring complexity, more electrode groups enable a lower achievable force ripple rate [see Fig. 2(i)]. For a fixed group number, tuning the phase offset θ_d effectively suppresses ripple [see Fig. 2(f)] and simultaneously improves $\min(F_p)$ [see Fig. 2(e)]. Further adjusting the group length provides an additional way to reduce ripple and increase $\min(F_p)$ [see Fig. 2(h)]. Reduced ripple improves displacement linearity, while a higher $\min(F_p)$ enhances the output force, thereby improving the overall actuator performance.

B. Finite Element Analysis

Following the analytical model prediction of the force ripple trend and the optimal phase-offset range, FEA was conducted to quantitatively evaluate the actuation force of the EFAs. In the simulation, the actuator was placed in an air-filled domain, with the outer boundaries modeled as infinite elements with zero charge. The slider adopted the offset electrode layout, while the stator used the straight layout. The electrode pitch, width, and spacing were set to 200, 100, and 100 μm, respectively. The material parameters of each layer are listed in Table I. A free tetrahedral mesh was employed in the simulations.

TABLE I
MATERIAL PARAMETERS OF THE ACTUATOR USED IN THE SIMULATION

Film	Thickness (μm)	Material	Relative permittivity
Electrode layer	12	Copper	–
Base layer	25	Polyimide	3.4
Cover layer	13	Polyimide	3.4
Adhesive layer	2	Acrylic acid	3.5
Air gap	20	Air	1

When the three-phase sinusoidal AC voltages were applied to the electrodes in the simulation, the electric potential distribution of the actuator was obtained, as shown in Fig. 3(a). The simulation result regarding the relationship between the peak force density per voltage squared and phase offset is presented in Fig. 3(b). The force ripple rate and $\min(F_p)$ reach the optimum at $\theta_d = 0.467\pi$ and $\theta_d = 0.400\pi$, respectively. Considering the tradeoff among electrode design parameters and actuator performance, $d = 125\,\mu$m ($\theta_d = 0.417\pi$) was selected for subsequent analysis.

The peak force versus electric angle at phase offset $\theta_d = 0.417\pi$ is shown in Fig. 3(c) with a ripple rate of 0.265 and a $\min(F_p)$ of $4.88 \times 10^{-4}\,\mathrm{N \cdot m^{-2} \cdot V^{-2}}$. A contour plot of the force characteristics at this phase offset is shown in Fig. 3(d), illustrating a smoother force ripple profile.

To assess the impact of fabrication tolerances, the electrode width was varied while maintaining a fixed electrode pitch of 200 μm. As shown in Fig. 3(e), reducing the electrode width to approximately 0.077 mm leads to a decrease of about 10% in $\min(F_p)$, while the force ripple rate is insensitive to width variations within this range. Based on these results, actuator films with an electrode width greater than 80 μm are considered acceptable.

III. FABRICATION

The fabrication process of the actuator film is depicted in Fig. 4(a)–(g). A double-sided copper-clad laminate, composed of a 25-μm polyimide base layer and two 12 μm copper layers, was employed as the substrate. Negative photoresist dry films were laminated onto both sides of the laminate [see Fig. 4(a)], followed by ultraviolet exposure through masks containing the predefined electrode layout [see Fig. 4(b)]. The irradiated photoresist regions were cross-linked to form insoluble hardened dry film, while the unexposed regions were dissolved in a 1% Na_2CO_3 developing solution [see Fig. 4(c)]. The exposed copper areas were subsequently etched using a $CuCl_2$/HCl etching solution [see Fig. 4(d)]. The remaining hardened photoresist was stripped with a 3% NaOH stripping solution [see Fig. 4(e)], leaving the patterned copper electrodes [see Fig. 4(f)]. For electrical insulation, two coverlays (each consisting of a 13-μm polyimide

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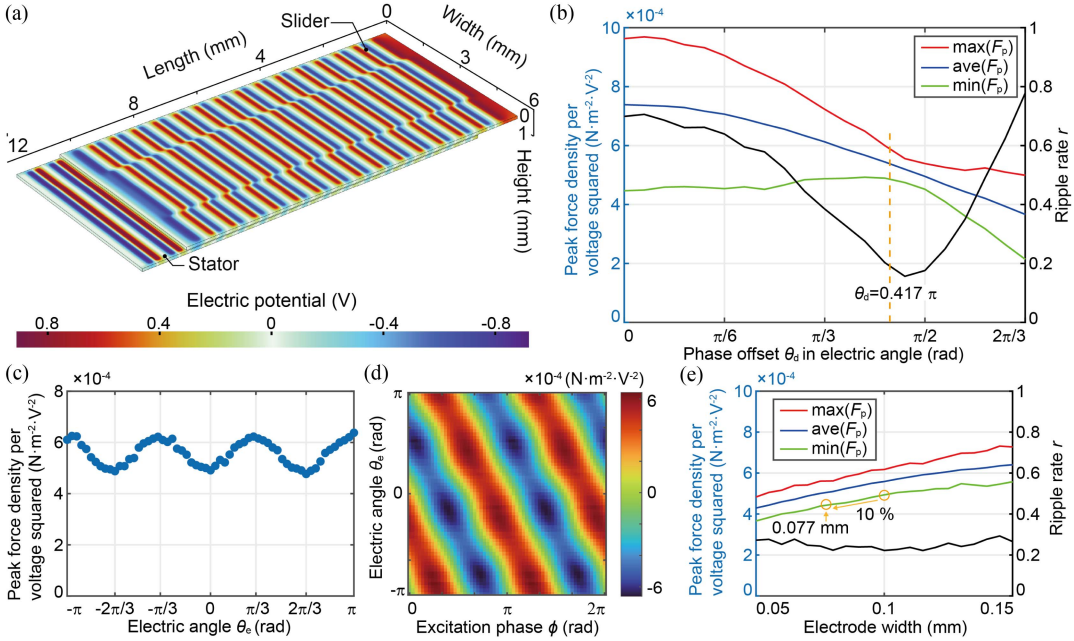


Fig. 3. FEA of the offset layout EFA. (a) Electric potential distribution [$x_{sl} = 1.2$ mm, $\phi = \pi$, and $\theta_d = 0.417\pi$ ($d = 125\,\mu$m)]. (b) Peak force density per voltage squared and ripple rate versus phase offset. (c) Peak force versus electric angle at phase offset $\theta_d = 0.417\pi$. (d) Contour plot for the force characteristics at phase offset $\theta_d = 0.417\pi$. (e) Effect of varying the electrode width on force at phase offset $\theta_d = 0.417\pi$.

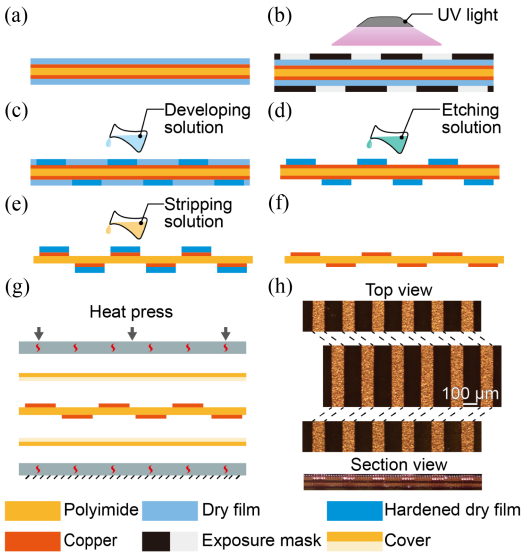


Fig. 4. Fabrication of the offset layout EFA. (a)–(g) Illustration of the fabrication process for the actuator film. (h) Electrode configurations in top and cross-sectional views.

layer and a 13-μm thermal adhesive layer) were bonded onto both sides of the patterned laminate through hot pressing at 185°C under 2.3 MPa for 3 h [see Fig. 4(g)].

The device-to-device variation of fabricated offset layout actuator films is summarized in Table II. The measured electrode widths are approximately 83.7 μm, slightly smaller than the designed 100 μm, which is likely attributed to overetching during fabrication. This deviation remains within an acceptable range and has a small impact on force performance. The actuator film had an overall size of 100 × 50 mm^2, an average thickness

TABLE II
DEVICE-TO-DEVICE VARIATION OF ACTUATOR FILMS

	Pitch (μm)	Offset (μm)	Width (μm)	Thickness (μm)
Sample 1	202	125	84.7	89
Sample 2	203	125	82.9	89
Sample 3	199	123	83.0	90
Sample 4	203	127	85.1	91
Sample 5	202	124	84.7	91
Sample 6	203	127	83.4	89
Sample 7	200	124	82.0	91
Mean ± SD	202 ± 1.5	125 ± 1.4	83.7 ± 1.2	90 ± 1

of 90 μm, and a mass of 0.86 g. The average electrode pitch and offset are 202 and 125 μm, respectively. The effective electrode area was 93.4 × 48.4 mm^2, including the bridge electrode region. Detailed electrode configurations are presented in the top and cross section views [see Fig. 4(h)].

IV. EXPERIMENTAL RESULTS

In this section, the offset layout actuator was experimentally evaluated in terms of displacement nonlinearity error, velocity, positioning performance, actuation force, force ripple, power consumption, and lifetime characterization. In these experiments, the control signals (three-phase sinusoidal voltages with a 120° phase shift) were generated by a controller board (SCB-68A, National Instruments) and amplified 1000-fold using three high-voltage amplifiers (Model 615-3, Trek) to drive the actuator. Sensor data were acquired through analog signal acquisition boards (TB-4309 and TB-4330, National Instruments). Data were acquired at 1000 Hz without additional filtering under

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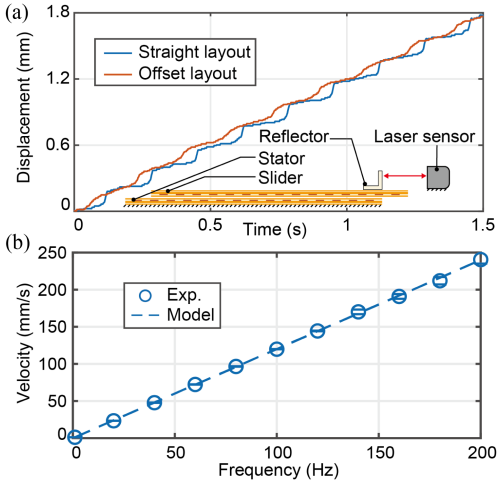


Fig. 5. Displacement and velocity of the actuator. (a) Experimental setup and displacement results at a driving frequency of 1 Hz. (b) Velocity results at different driving frequencies. Error bars represent mean ± 1 SD ($n = 3$).

laboratory conditions (ambient temperature 20°C–25°C and relative humidity 40%–60%).

A. Motion Performance

The displacement test platform is shown in Fig. 5(a). The stator of the actuator was fixed onto an optical table, while a reflector was mounted on the slider to measure its real-time position using a laser displacement sensor (HG-C1050, Panasonic, range: ±15 mm, linearity: ±0.1% full scale, and repeatability: 30 μm). To evaluate the displacement nonlinearity error, the actuator was driven by sinusoidal voltages with $V_{0\text{-p}} = 500$ V at a frequency of 1 Hz. The displacement results of the offset layout and straight layout sliders are presented in Fig. 5(a), showing that the displacement curve of the offset layout actuator is noticeably smoother. Considering a full scale of 50 mm, the offset layout slider achieved a nonlinearity error (using the best-fit-line method) of 0.115% of full scale, representing a 13.5% reduction compared to the straight layout slider (0.133% of full scale), demonstrating superior linearity characteristics.

The velocity results under different frequencies are shown in Fig. 5(b). The velocity increased approximately linearly with the driving frequency and reached a maximum of about 241 mm/s at 200 Hz. The relationship between velocity and frequency generally agrees with the model derived from (1).

For positioning performance tests, the offset layout actuator was driven by voltages with $V_{0\text{-p}} = 500$ V at a frequency of 10 Hz. Continuous commanded positioning to a target displacement of 20 mm was performed, as shown in Fig. 6(a). Stepwise positioning at discrete target points with 4 mm intervals, together with the corresponding positioning residuals, is shown in Fig. 6(b) and (c). The envelope of positioning residuals remains within 0.276 mm over the full 20 mm displacement range.

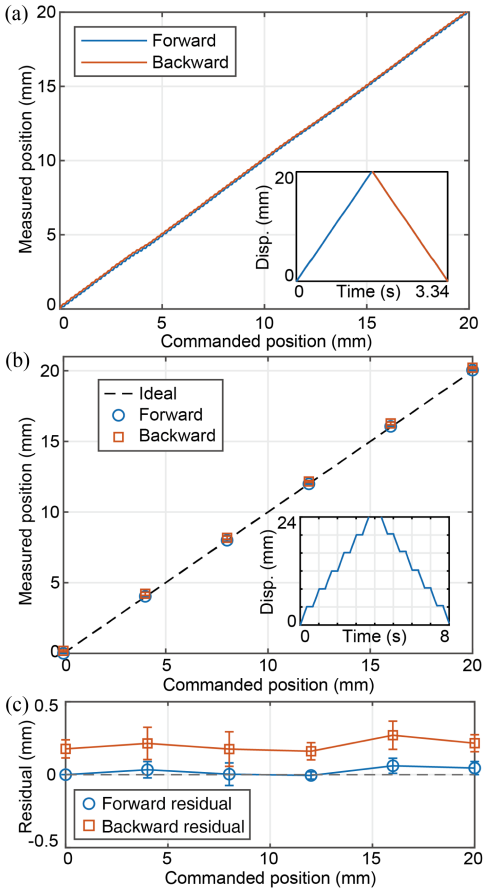


Fig. 6. Positioning performance of the offset layout actuator. (a) Continuous forward and backward measured trajectories. (b) Stepwise positioning at discrete target points, illustrating bidirectional hysteresis. (c) Residuals for forward and backward motions. Error bars represent mean ± 1 SD ($n = 3$).

B. Actuation Force Characteristics

The experimental platform for actuation force measurement is shown in Fig. 7(a). The stator was mounted on a sliding rail fixed to an optical table using screws. The slider was connected by a hook to a spring attached to a force sensor. During the actuation force tests, the slider was driven by sinusoidal voltages with varying amplitudes at a frequency of 10 Hz. As the slider stretched the spring leftward, the measured force gradually increased until the slider could no longer extend the spring, and this peak force was recorded as the output force corresponding to the given voltage amplitude. The relationship between output force and voltage amplitude is plotted in Fig. 7(a). The force increased quadratically with the voltage amplitude. When $V_{0\text{-p}} < 500$ V, the experimental results were lower than those predicted by FEA, likely due to intrinsic deformation of the film.

The offset actuator achieved a maximum output force of approximately 0.911 N at $V_{0\text{-p}} = 600$ V, representing an improvement of about 22.3% compared with the straight layout (0.745 N). The offset layout actuator also achieved a force density of up to 183 N/m^2, corresponding to a 44.1% improvement over previously reported three-phase EFAs operating in air, as

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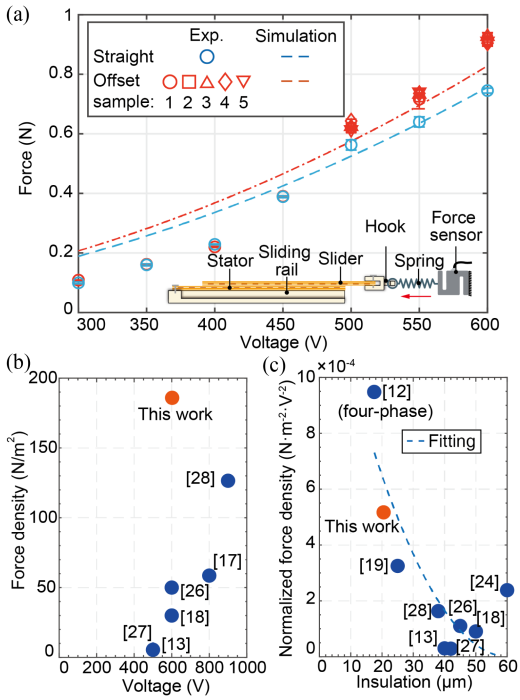


Fig. 7. Actuation force of the actuator. (a) Experimental setup and force measurements under different voltage amplitudes at a driving frequency of 10 Hz. (b) Comparison of force density with three-phase EFAs operating in air. (c) Comparison of force density per voltage squared with EFAs. Error bars represent mean ± 1 SD ($n = 5$).

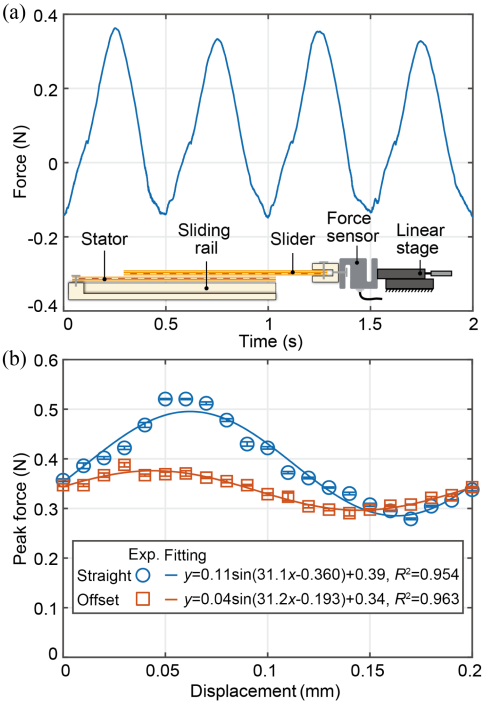


Fig. 8. Force ripple of the actuators. (a) Experimental setup and force measurement for the offset slider at a displacement of 0.05 mm under a 1-Hz driving frequency. (b) Comparison of the peak actuation force between straight and offset electrode layouts. Error bars represent mean ± 1 SD ($n = 5$).

TABLE III
COMPARISON OF PREVIOUS EFAS

Ref.	Electrode layout	Voltage (V)	Pitch (μm)	Insulation (μm)	Force density ($\mathrm{N \cdot m^{-2}}$)	Normalized force density*
[18]	Straight	600	400	50	∼30	0.833
[26]	Straight	600	300	45	∼50	0.916
[13]	Straight	500	400	40	5.74	0.230
[27]	Straight	500	400	42	5.45	0.218
[28]	Straight	900	300	38	127	1.56
[20]	Skew	1200	200	–	396	2.75
[24]	Skew	1400	300	60	453	2.31
[19]	Straight	1300	200	25	537	3.18
[12]	Straight (four-phase)	500	67	17.5	237	9.48
This work	Offset	600	200	20.5	183	5.09

* Normalized force density is expressed in $\times 10^{-4}\,\mathrm{N \cdot m^{-2} \cdot V^{-2}}$.

summarized in Fig. 7(b) and Table III. A further comparison of the force density per voltage squared with previously reported EFAs operating both in air and in dielectric liquids is shown in Fig. 7(c), where the offset actuator exhibits a relatively high performance. Although a four-phase EFA with a smaller electrode pitch and thinner insulation can achieve higher normalized force density, it requires an additional amplifier and more complex fabrication processes.

For ripple rate measurements, the slider, force sensor, and linear stage (LX125-C, straightness of $\leq 0.02\,\mathrm{mm}/25\,\mathrm{mm}$, minimum step size of 0.01 mm) were sequentially fixed, as illustrated in Fig. 8(a). The actuator was driven by sinusoidal voltages with $V_{0\text{-p}} = 400\,\mathrm{V}$ at a frequency of 1 Hz. At the beginning of the test, the slider and stator were positioned to maximize their overlapping area, which was defined as the displacement origin (slider displacement relative to the stator equal to 0 mm). The output force measured at a displacement of 0.05 mm is shown in Fig. 8(a), where the peak value of the force curve was recorded as F_p. After each measurement, the linear stage was incrementally shifted by 0.01 mm. This procedure yielded the relationship between peak force and displacement of sliders, exhibiting clear periodicity ($T = 2\pi/31.1\,\mathrm{mm} \approx 0.202\,\mathrm{mm}$), as shown in Fig. 8(b). The measured force ripple rate of the offset layout actuator is 0.290, which is close to the FEA predicted value of 0.265 and represents a 53.4% reduction compared with the straight layout (0.622). Under the present actuator scale and driving conditions, $\min(F_\mathrm{p})$ was 0.279 N (straight) and 0.291 N (offset), while the force variation $(\max(F_\mathrm{p}) - \min(F_\mathrm{p}))$ reduced from 0.242 N (comparable to $\min(F_\mathrm{p})$) to 0.0974 N, indicating larger and noticeably smoother output force characteristics.

C. Power Consumption

The experimental setup for output power measurement is shown in Fig. 9(a). The slider was connected to a hook and further linked by a string to a cup loaded with small steel beads, which were used to realize different payload masses. The actuator was driven by sinusoidal voltages with $V_{0\text{-p}} = 500\,\mathrm{V}$ at 10 Hz. The slider was given approximately 10 mm of leftward motion space for lifting the payload. Representative feedback

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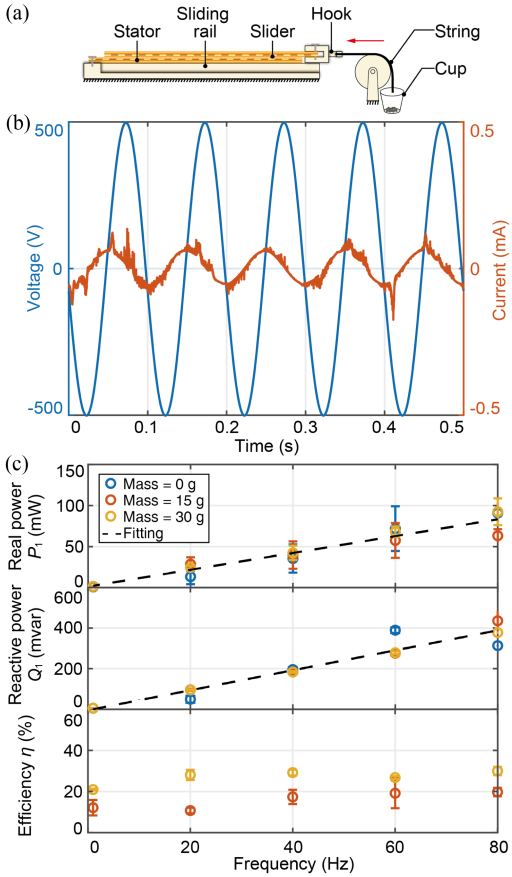


Fig. 9. Power consumption of the actuator. (a) Experimental setup. (b) Voltage and current signals of the high-voltage amplifier while the slider lifts the cup under a 10-Hz driving frequency and a 15-g payload. (c) Real power, reactive power, and efficiency across different frequencies and payloads. Error bars represent mean ± 1 SD ($n = 3$).

voltage and current signals from the high-voltage amplifier under a 10 Hz driving frequency and a 15-g payload (mechanical output power $P_{\text{out}} = 1.76$ mW) are shown in Fig. 9(b), the average electrical input power P_{ave} was calculated to be about 11.73 mW. The real power P_1 (11.73 mW) and reactive power Q_1 (46.8 mvar) were calculated via *I–V* phase analysis. Accordingly, the energy conversion efficiency $\eta = 100\% \times P_{\text{out}}/P_{\text{ave}}$ was determined to be approximately 15.0%. The results in Fig. 9(c) show that varying the payload mass (0 g, 15 g, and 30 g) has only a minor influence on both P_1 and Q_1, while both increase approximately linearly with driving frequency. The efficiency increases with payload mass but exhibits no clear monotonic trend with frequency within the tested range.

D. Lifetime Characterization

As shown in Fig. 10(a), the actuator can operate for more than 10000 cycles with $V_{0\text{-p}} = 500$ V at 10 Hz and a stroke of 10 mm while maintaining stable reciprocating motion without observable degradation in displacement amplitude. To assess force degradation, the method shown in Fig. 8(a) was employed with $V_{0\text{-p}} = 500$ V at 100 Hz. As shown in Fig. 10(b), the peak output force decreases gradually with increasing cycle number. In addition, optical images of the electrodes were compared

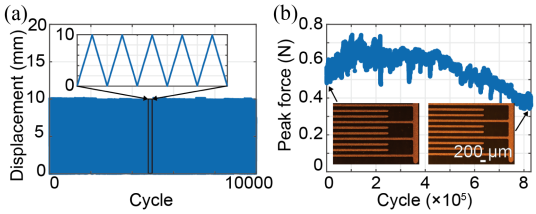


Fig. 10. Lifetime characterization of the actuator at $V_{0\text{-p}} = 500$ V. (a) Cycling test under a 10-mm stroke at a 10-Hz driving frequency. (b) Cycling test showing peak force degradation at a 100-Hz driving frequency.

before testing and after 8×10^5 cycles, and no obvious macroscopic defects, such as void formation, were observed. Nevertheless, given the force reduction, subtle degradation, such as dielectric charging or electrode migration, cannot be ruled out. These effects will be systematically investigated in the future.

V. Demonstrations

The offset layout actuator features an ultrathin, flexible, and lightweight structure, while exhibiting small displacement nonlinearity error, high output force, small force ripple, and low power consumption. These advantages enable the actuator to be applied in a broader range of applications.

A. Lens Focusing Module

Current lens focusing techniques typically employ voice coil motors (VCMs) [29], [30], [31] or piezoelectric actuators [32], [33], [34]. Magnet-based actuators suffer from large supply currents, and the heat generated during long-term operation may cause demagnetization and degrade stability [35], [36]. Piezoelectric actuators are costly and provide only a limited motion stroke [37]. With a thin structure, sufficient motion linearity, a large motion stroke, and low power consumption, the proposed offset actuator can be employed for lens focusing.

As shown in Fig. 11(a), the stator was mounted on a sliding rail, and a lens (20.2 g, approximately 11.7 times the actuator weight) was attached to the left end of the slider. A smartphone camera (IMX890, Sony) was fixed on the left side of the lens, while a target object printed with the word "Focus" was fixed on the right side of the lens. The camera was initially focused on the lens surface, as shown in the camera view at 0 s. The actuator was then driven by sinusoidal voltages with $V_{0\text{-p}} = 500$ V at a driving frequency of 10 Hz. As the slider moved the lens rightward, the camera view gradually became clearer due to the adjustment of the optical path, reaching optimal focus at approximately 2.3 s, as shown in the movie given in the Supplementary Material.

The settling time and repeatability of the proposed actuator were evaluated at a representative displacement of 10 mm. The corresponding settling time and repeatability ($n = 3$) at a driving frequency of 100 Hz were approximately 85 ms and 56 μm, respectively, as shown in Fig. 11(b). The settling time was averaged over three measurements (91, 82, and 83 ms), while the repeatability was defined as the maximum variation among the measured final positions (10.073, 10.017, and 10.052 mm). A

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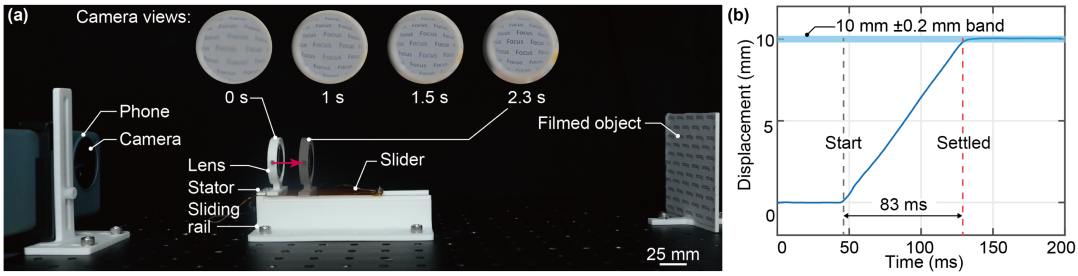


Fig. 11. Lens focusing demonstration. (a) Slider drove the lens rightward to adjust the optical path and achieved a clear focus under a 10-Hz driving frequency. (b) Settling time of the lens to achieve the target position of 10 mm under a 100-Hz driving frequency.

TABLE IV
COMPARISON OF LENS FOCUSING ACTUATORS

Ref.	Type	Settling time (ms)	Repeatability (μm)	Stroke (mm)	Length (mm)	Width (mm)	Thickness (mm)
[29]	VCM	39	–	3	41.5	41	6.3
[30]	VCM	8	5	0.5	6.5	6.5	4
[31]	VCM	10	2	0.3	6.5	6.5	4
[32]	Piezo	10	–	0.02	∼8	∼8	–
[33]	Piezo	2.5	–	0.01	∼10	∼10	–
[34]	Piezo	116	–	0.06	∼8	∼8	–
This work	EFA	85	56	10	50	0.195	100

comparison with VCM-driven and piezo-driven focusing modules is provided in Table IV. The proposed electrostatic actuator primarily demonstrates advantages in achieving a long stroke and occupying minimal lateral volume around the lens, making it suitable for applications that require large lens travel. Further improvements in repeatability are expected through enhanced control strategies in future work.

B. Ultrathin Retractable Camera Module

In recent years, front-facing cameras in smartphones have often adopted punch-hole designs, which improve the screen-to-body ratio while maintaining a compact form factor, but inevitably compromise display integrity. Some smartphone manufacturers previously introduced retractable front cameras to achieve a full-screen display. However, the large volume and mass of the actuation modules resulted in this solution being gradually phased out. In contrast, the EFA possesses a lightweight and ultrathin profile, offering significant potential for integration in compact devices.

The present EFA consists of two separate slider and stator films. When the actuator was oriented vertically, the flexible structure may bend or deform, causing the two films to peel apart and preventing operation. Therefore, packaging is necessary to ensure reliable operation in the vertical orientation. As illustrated in Fig. 12(a), a polyimide film ($102 \times 54\,\mathrm{mm}^2$) and a stator ($100 \times 50\,\mathrm{mm}^2$) were bonded together using two strips of polyimide tape to form a flexible package with a central cavity. The slider ($100 \times 48\,\mathrm{mm}^2$) was inserted into the cavity, where its smaller width than that of the stator allowed free movement without interference from the tapes. The resulting actuation module measured only 0.257 mm in thickness and 2.17 g in mass. A micro camera (OV9734, OmniVision Technologies Inc.) and data cable were attached to the top of the slider using polyimide tape, and the module was mounted onto the smartphone through a 3D-printed slot.

To quantify the drag force, the slider was connected to a force sensor (LSB203, MTS) and pulled at a constant velocity of 50 mm/min using a tensile testing platform (Model 42, MTS). As shown in Fig. 12(b), the drag force increases with displacement in both cases, and the addition of the camera module produces a higher drag force. The actuator was driven by sinusoidal voltages with $V_{0\text{-p}} = 500$ V at a driving frequency of 10 Hz. The camera was driven in a retractable motion by the slider, with its direction alternating every 1.5 s, as shown in Fig. 12(c) and in the movie given in the Supplementary Material. At 0 s, the camera was hidden inside the slot and had no external view. After 0.5 s of motion, the camera extended and began to capture a partial view, and by 1.5 s, a complete view was displayed on the screen.

The voltage and current signals of the amplifier while driving the actuator are shown in Fig. 12(d). Two larger current spikes, separated by approximately 1.5 s, can be observed and are caused by voltage transitions during direction reversal. The input power at different driving frequencies is plotted in Fig. 12(e) and increases approximately linearly with frequency.

C. Flip Camera Module

Contemporary mobile smartphones are typically equipped with both front and rear camera modules. The integration of two independent modules increases cost and bulk. Moreover, the limited space available for the front camera often results in a lower resolution than the rear one, which reduces the quality of front-facing images. A flip camera smartphone addresses this issue by allowing a single camera module to be rotated, either passively or actively, to any desired position. This not only enhances the pixel quality of front-facing images but also provides greater flexibility in shooting angles. However, passive flipping requires manual operation, whereas active flipping is commonly driven by electromagnetic motors, resulting in a heavier and bulkier structure. In contrast, the offset layout EFA, with its ultrathin and lightweight structure, shows great potential for driving flip camera modules.

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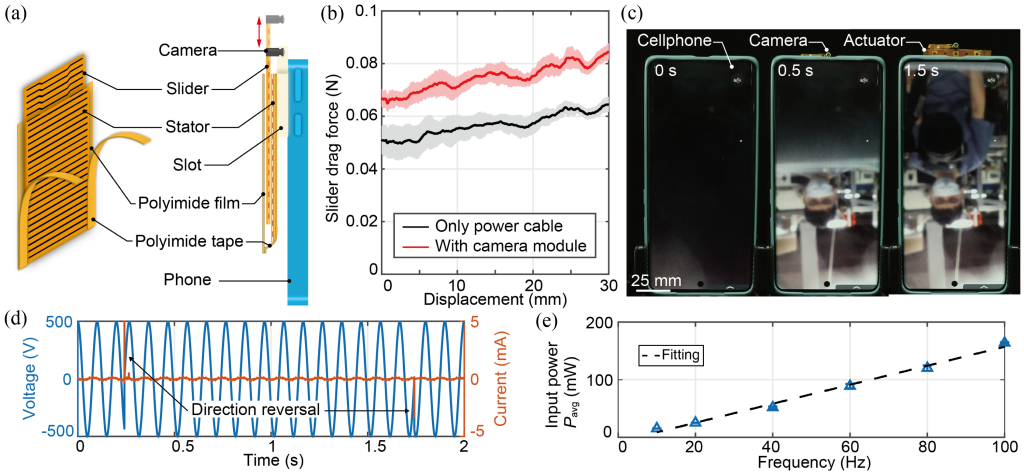


Fig. 12. Ultrathin retractable camera module. (a) Schematic of the packaged actuation module and its integration on the smartphone. (b) Drag force of the slider with only the power cable and with the camera and data cable. (c) Retractable camera extending motion with the corresponding smartphone screen display. (d) Voltage and current signals of the amplifier while the slider operates at a 10 Hz driving frequency. (e) Input power of the actuation module (camera excluded). Confidence bands ($n = 5$) and error bars ($n = 3$) represent mean $\pm$ 1 SD.

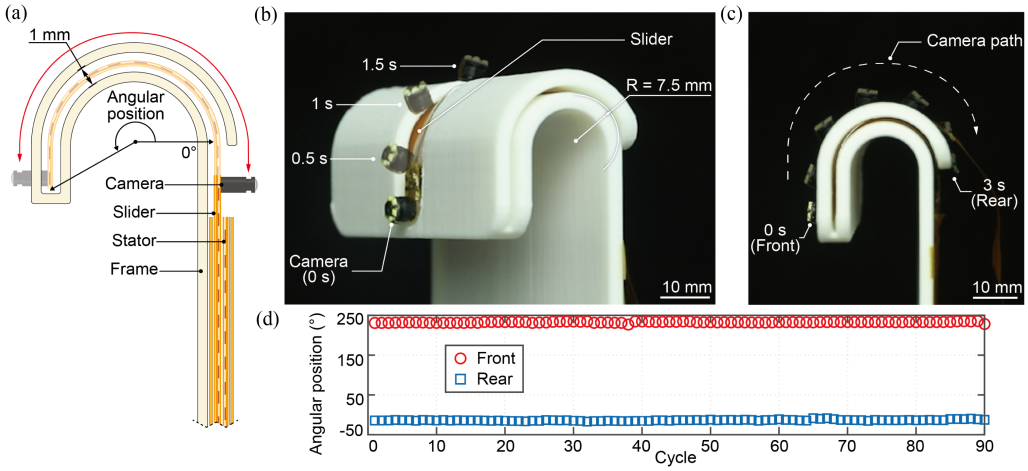


Fig. 13. Flip camera module. (a) Schematic of the module structure and flipping process, where the slider adaptively bends to match the frame shape during motion. (b) Oblique and (c) side views of the slider driving the camera to flip. (d) Angular positions of the front and rear points over repeated cycles.

As shown in Fig. 13(a), the module consists of a frame and the packaged actuation module with the camera. The upper part of the frame features a semicircular slit with an inner radius of 7.5 mm and a width of 1 mm, with an additional 10 mm wide opening at its center to allow camera passage. During operation, the slider carrying the camera adaptively bends to match the frame shape. The actuator was driven by voltages with $V_{0\text{-p}} = 500$ V at 10 Hz. The slider drove the camera through a flipping motion as shown in the movie given in the Supplementary Material, with its direction alternating every 3 s, moving the camera along a semicircular path with a radius of 7.5 mm, as shown in Fig. 13(b) and (c).

At 0 s, the camera was in the front position, and after 3 s of motion, it was brought to the rear position, thereby enabling smooth switching. As defined in Fig. 13(a), the flipping motion is quantified by the angular position of the slider. The angular positions of the front and rear points over repeated reciprocating cycles are shown in Fig. 13(d). Under the tested condition (bend radius $R = 7.5$ mm), the actuation module performed approximately 90 reciprocating cycles (from about $-13°$ to 233°) without obvious degradation.

D. Circumferential Wide-Range Camera Module

Most cameras have a fixed field of view, to expand the camera's viewing range, a separate rotary base, such as that used in pan–tilt cameras, is often incorporated, which inevitably increases the overall volume and mass and complicates system integration [38], [39]. The EFA, owing to its flexible structure, can maintain motion even under bending or deformation. This feature enables the actuator to be easily deployed on planar or curved surfaces to carry a camera enlarging its viewing range, realizing a wide-range camera module.

To extend the travel range of the slider, we designed an elongated packaged actuation module, as illustrated in Fig. 14(a). The module consists of two stators (200 $\times$ 54 mm^2 in total) and

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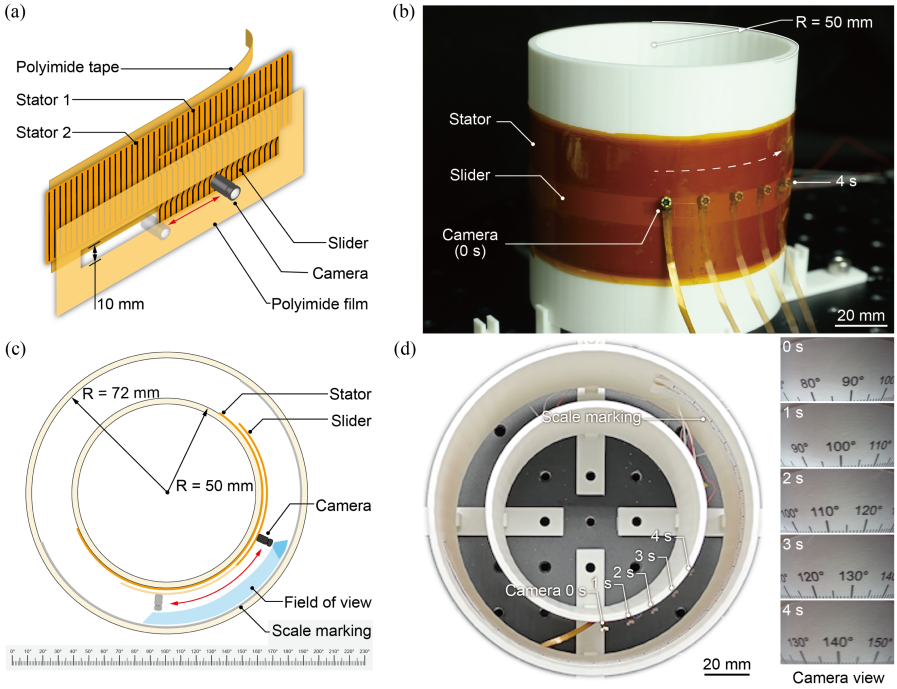


Fig. 14. Circumferential wide-range camera module. (a) Schematic of the packaged actuation module with two stators and one slider. (b) Motion of the camera module attached to the outer surface of a cylinder with a 50-mm radius. (c) Schematic of the camera module applied to scan the scale marking on the inner wall of a cylinder with a 72-mm radius. (d) Scanning process, where the camera captures about 42° of the inner wall when stationary, while slider motion increases the accessible viewing angle to up to 93°.

one slider ($100 \times 48\,\text{mm}^2$). A polyimide film ($180 \times 54\,\text{mm}^2$) with a central slot ($140 \times 10\,\text{mm}^2$), together with two polyimide tapes, was used to encapsulate the stators and the slider to form the actuation module. The camera was attached to the slider through the slot using polyimide tape. As shown in Fig. 14(b), the actuation module was mounted on the outer wall of a cylinder with a radius of 50 mm. The actuator was driven by voltages with $V_{0\text{-p}} = 500\,\text{V}$ at 10 Hz. The slider carried the camera in a retractable motion, with its direction alternating every 4 s, moving the camera along a circular path to realize the circumferential wide-range camera, as demonstrated in the movie given in the Supplementary Material.

Furthermore, as shown in Fig. 14(c), an additional cylinder with an inner wall radius of 72 mm was placed outside, with scale markings from 0° to 230° corresponding to the circumferential angle of the inner wall. The slider drove the camera in circumferential motion, thereby expanding the accessible viewing range around the cylinder. At 0 s, the camera captured the region between 65° and 107° (an accessible viewing range of 42°), whereas after 4 s of motion the visible region extended to around 158° (an accessible viewing angle of 93°), representing a 121% increase in viewing range relative to the static configuration (see Fig. 14(d) and movie given in the Supplementary Material). These results demonstrate that the module enables wide-range imaging through actuator-driven camera translation.

VI. Conclusion

This article has presented a novel EFA with an electrode offset layout that effectively addresses the limitations of conventional straight electrode EFAs. By introducing a spatial phase offset, the proposed layout design suppressed force ripple by about 53.4% (0.622–0.290) while simultaneously enhancing output force by 22.3% (0.745–0.911 N). Demonstrations in driving lens focusing, smartphone retractable camera, flip camera, and circumferential wide-range camera modules highlight the actuator's potential for integration into compact optical systems. The next step is to integrate the proposed actuator into a fully functional smartphone prototype to drive a camera module.

Future work will extend the analytical model to incorporate bridge and fringing effects for further performance enhancement. In parallel, the sensitivity of the lubrication strategy and related improvement approaches [40], together with air-breakdown probability and performance degradation under varying environmental conditions (e.g., humidity, altitude, and surface contamination), will be investigated through detailed experiments and quantitative analysis. Practical deployment will require appropriate high-voltage protection and compliance with relevant safety and electromagnetic compatibility (EMC)/electromagnetic interference (EMI) standards, followed by integration into complete optical systems for comprehensive system-level performance evaluation. Overall, the electrode offset layout offers a versatile strategy for advancing both the performance and practical applicability of EFAs.

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